ML Detection Engine: Implementation and Evaluation Report

1. Architecture Alignment and Strategy

To satisfy the zero-trust requirements of the Dynamic Orchestration Framework, the ML Detection Engine was designed using a dual-layered approach. Recognizing that a single global model would generate high false positive rates (FPR) across diverse 5G traffic, the engine was heavily decentralized. Separate models were trained exclusively for each slice type (eMBB, URLLC, mMTC) using the 5G-SATORI dataset, ensuring that the detection boundaries accurately reflected the unique traffic behaviors of each network slice.

2. Data Preprocessing and Feature Engineering

To ensure low-latency real-time inference within the VNF sidecar monitors, the raw 5G-SATORI telemetry underwent aggressive dimensionality reduction and cleaning:

- **Data Sanitization:** Artifacts such as empty trailing columns and missing labels were stripped. Infinite values resulting from division-by-zero flow calculations were converted to zero.
- **Zero-Variance Filter:** Features that remained static across all flows were dropped.
- **Correlation Matrix:** A strict correlation threshold (> 0.90) was applied to strictly benign traffic to eliminate redundant network features, drastically reducing the computational weight of the final models.
- **Standardization:** The remaining highly independent features were scaled using StandardScaler to prevent high-magnitude features (e.g., total byte volume) from dominating the machine learning algorithms.

3. Unsupervised Learning: Zero-Day Anomaly Detection

To detect novel, unseen attacks that deviate from normal slice behavior, two unsupervised models were trained exclusively on Benign traffic:

- **Packet-Level Detection (One-Class SVM):** An SVM utilizing a Radial Basis Function (RBF) kernel was deployed to draw a hard decision boundary around normal traffic. Evaluation showed an exceptionally high benign recall rate (averaging 97-98%), proving its safety for the network. However, its malicious recall was lower (e.g., 34% for eMBB), as it evaluated traffic packet-by-packet and was easily bypassed by stealthy, legitimate-looking packets used in Request Bombing.
- **Flow-Level Temporal Detection (LSTM Autoencoder):**

  To catch the sequential build-up of DoS and QoS Degradation attacks, an LSTM Autoencoder was engineered to process traffic in a sliding window of 10 time steps. By measuring the Mean Squared Error (MSE) of the sequence reconstruction, the LSTM successfully flagged temporal anomalies that the SVM missed.
- **Dynamic Statistical Thresholding:** Instead of relying on a static error threshold, the system mathematically calculated the 95th percentile of the reconstruction error from the benign training data. Slice-specific thresholds were established (eMBB: 0.9177, URLLC: 2.8485, mMTC: 1.1438). This dynamic tuning successfully capped the False Positive Rate at exactly 5%, protecting healthy VNFs from unnecessary isolation while maintaining strict security.

4. Supervised Learning: Known Threat Classification

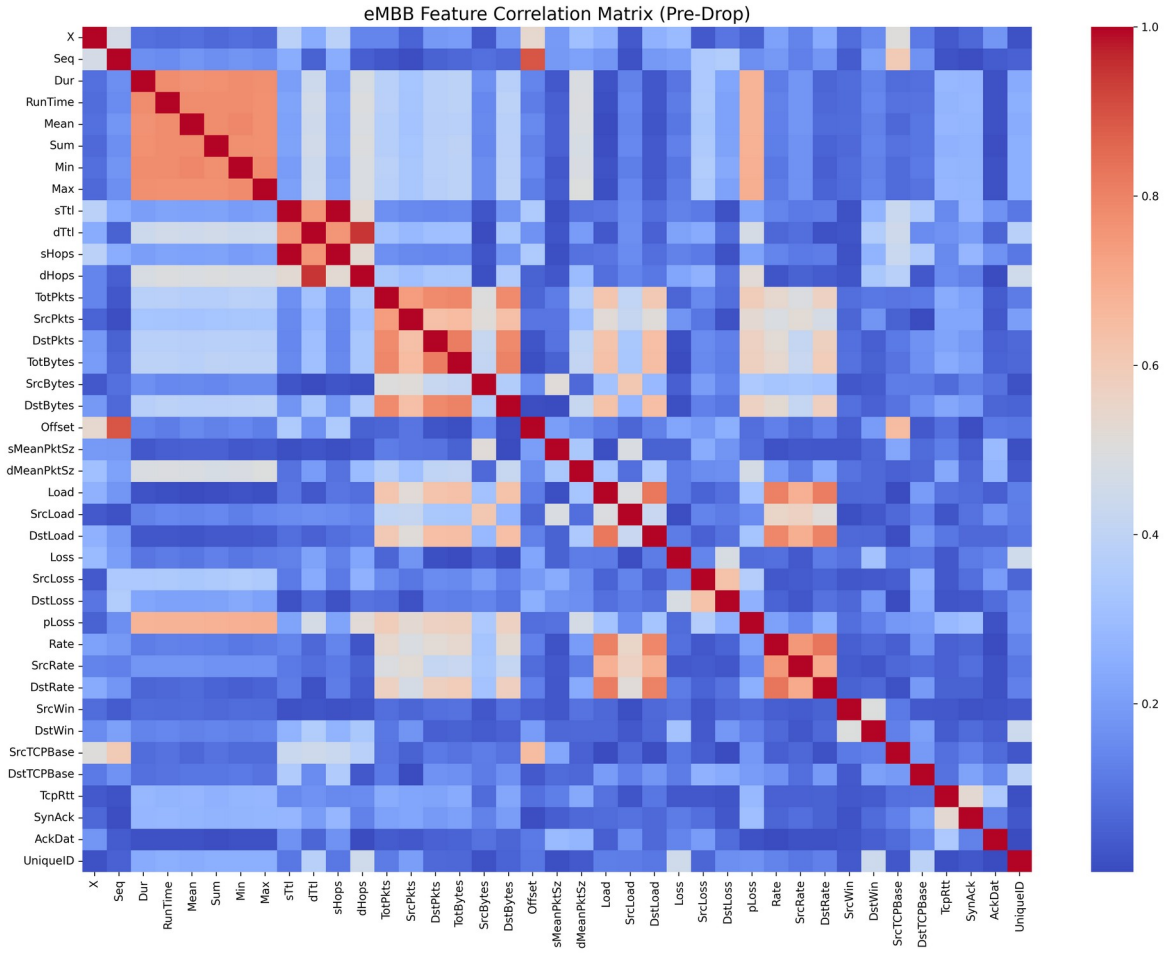
To complement the zero-day anomaly detectors, a supervised learning layer was introduced to perfectly identify known attack vectors.

- **XGBoost Classifier:** An Extreme Gradient Boosting model was trained on a mixed dataset containing both benign traffic and simulated attack traffic.

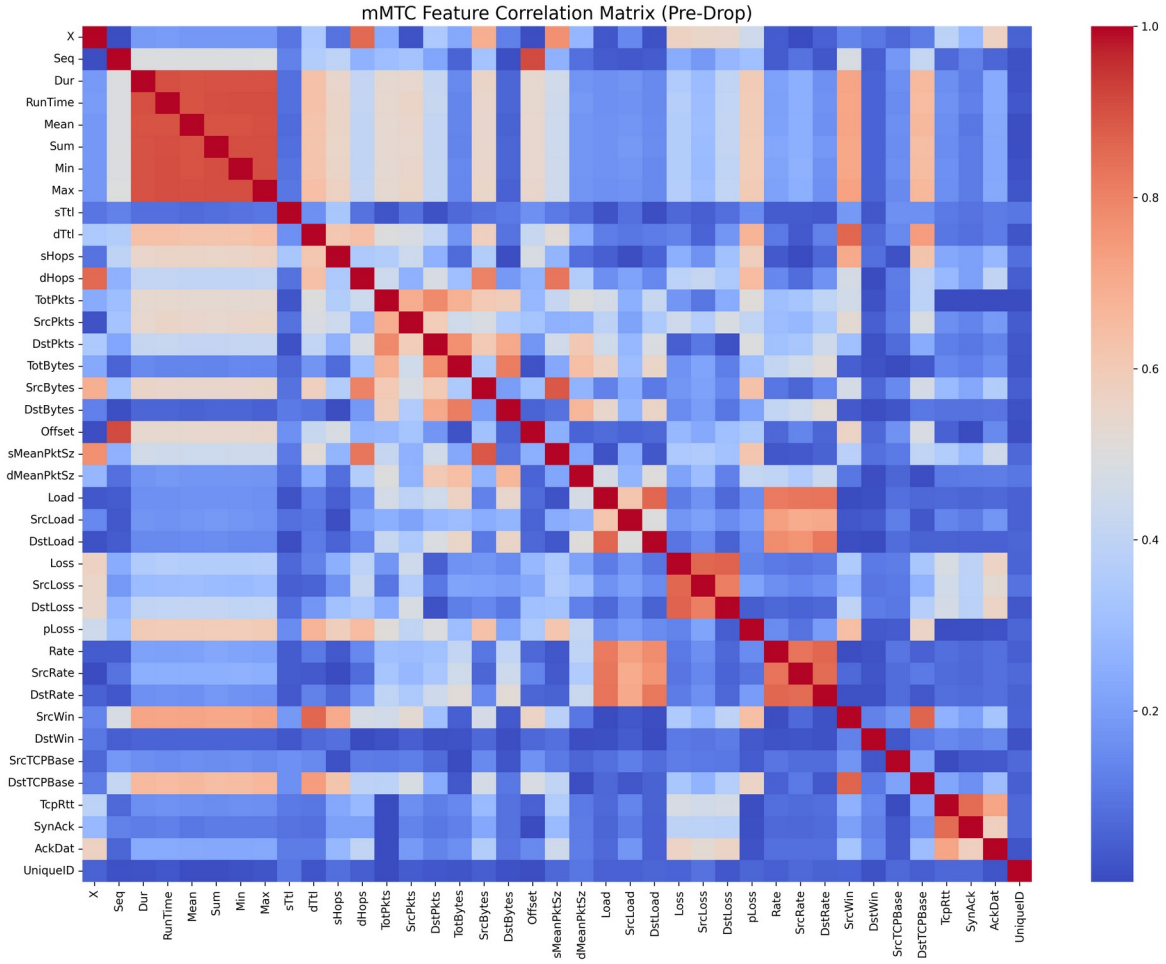
- **Results:** By memorizing the exact mathematical signatures of the Request Bombing and QoS Degradation scripts, the XGBoost model achieved a flawless **1.00 (100%) Precision, Recall, and F1-Score** across all three slices on the 20% testing holdout. This provided an impenetrable frontline defense against known threats.

Correlation Matrix

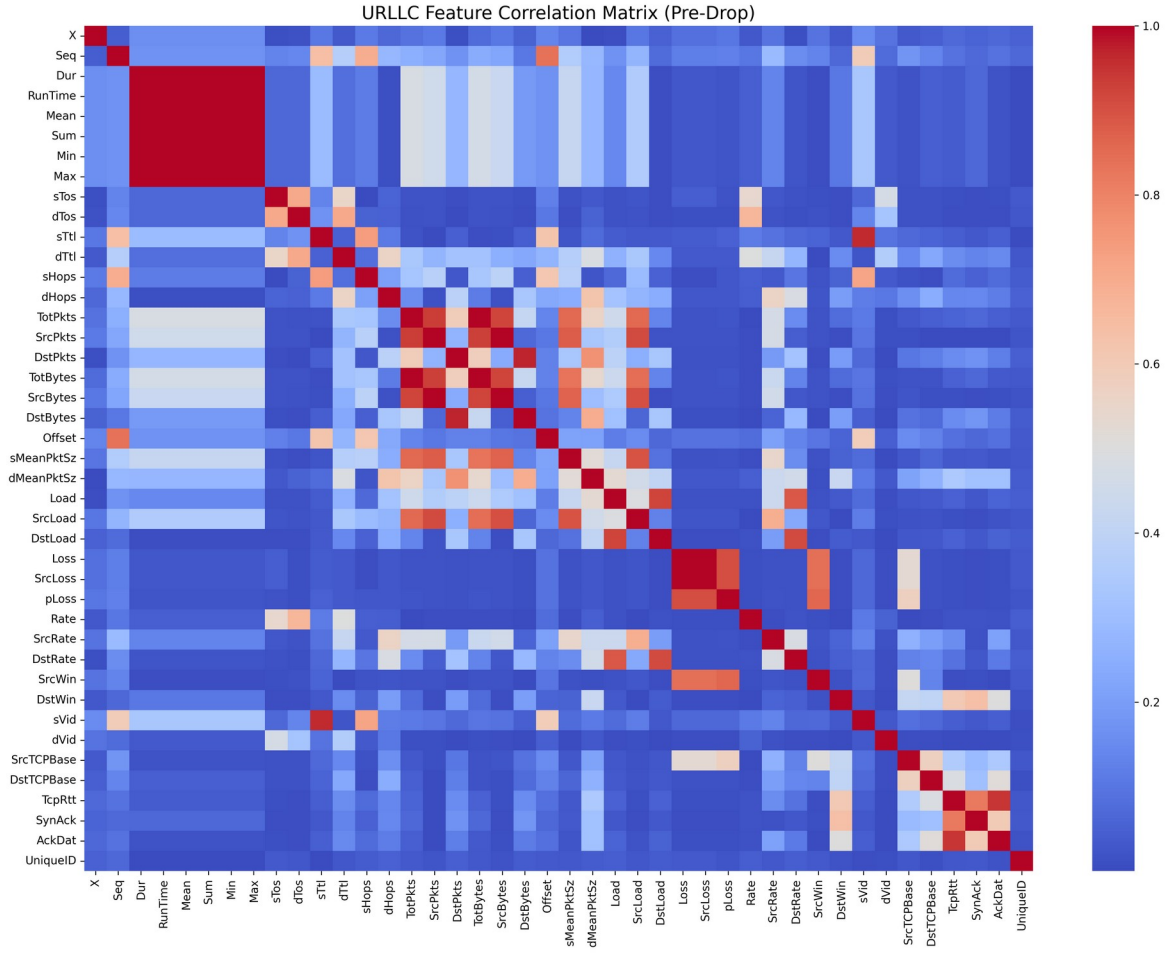
1. eMBB



2. mMTC

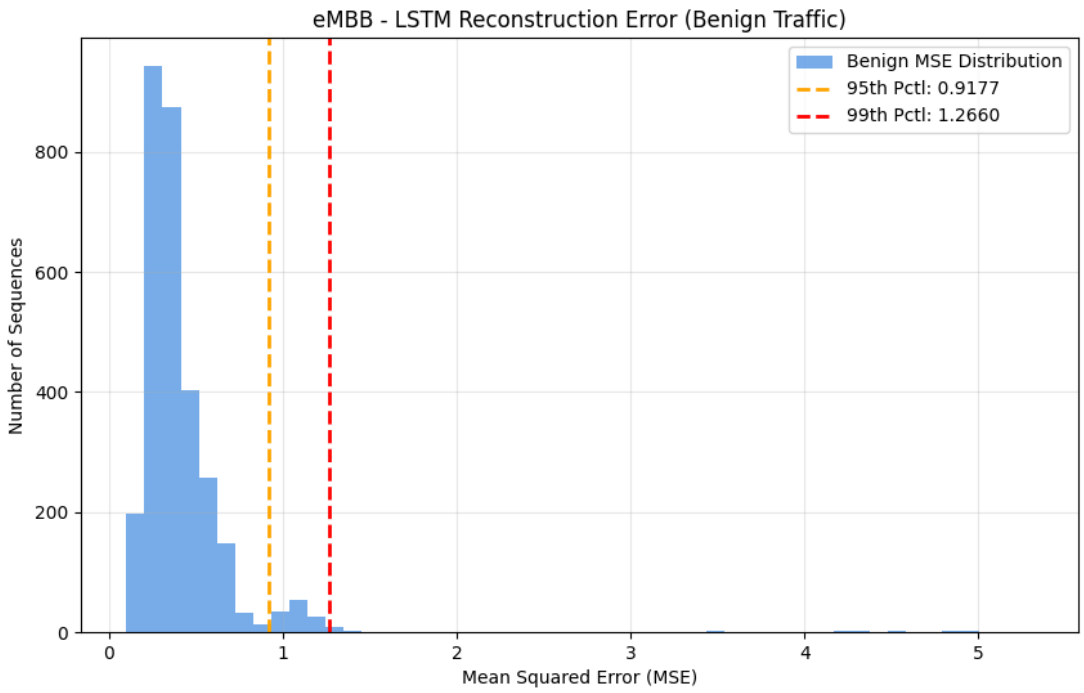

mMTC Feature Correlation Matrix (Pre-Drop)
X
Seq
Dur
RunTime
Mean
Sum
Min
Max
sTtl
dTtl
sHops
dHops
TotPkts
SrcPkts
DstPkts
TotBytes
SrcBytes
DstBytes
Offset
sMeanPktSz
dMeanPktSz
Load
SrcLoad
DstLoad
Loss
SrcLoss
DstLoss
pLoss
Rate
SrcRate
DstRate
SrcWin
DstWin
SrcTCPBase
DstTCPBase
TcpRtt
SynAck
AckDat
UniqueID
1.0
0.8
0.6
0.4
0.2

3. uRLLC

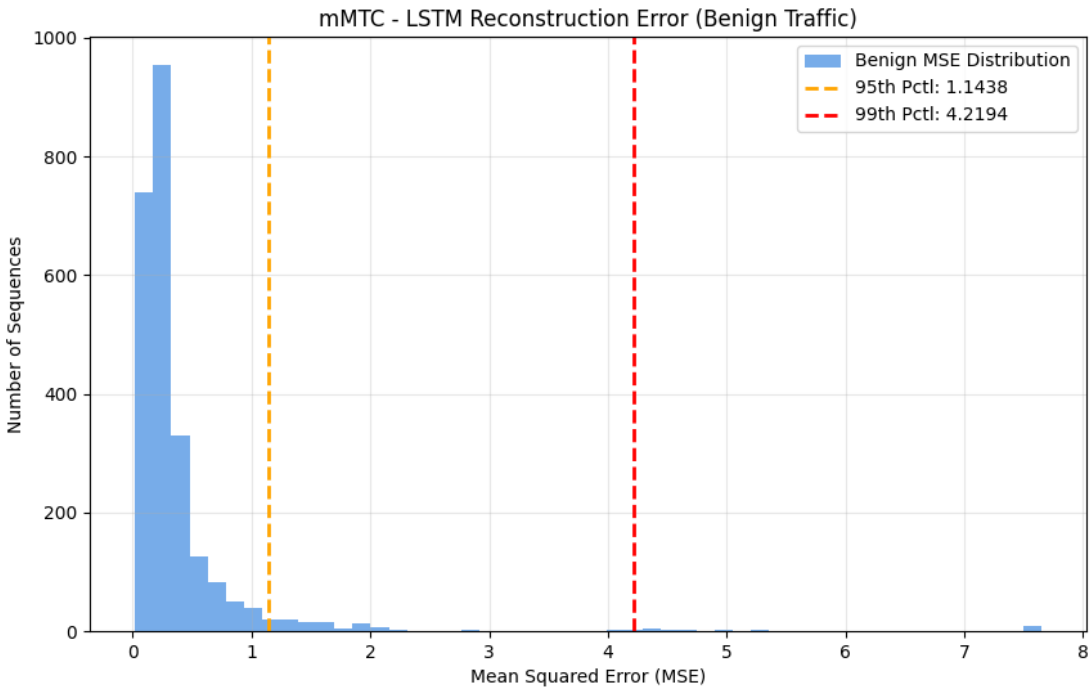

URLLC Feature Correlation Matrix (Pre-Drop)
X
Seq
Dur
RunTime
Mean
Sum
Min
Max
sTos
dTos
sTtl
dTtl
sHops
dHops
TotPkts
SrcPkts
DstPkts
TotBytes
SrcBytes
DstBytes
Offset
sMeanPktSz
dMeanPktSz
Load
SrcLoad
DstLoad
Loss
SrcLoss
pLoss
Rate
SrcRate
DstRate
SrcWin
DstWin
sVid
dVid
SrcTCPBase
DstTCPBase
TcpRtt
SynAck
AckDat
UniqueID
1.0
0.8
0.6
0.4
0.2

LSTM Threshold calculation

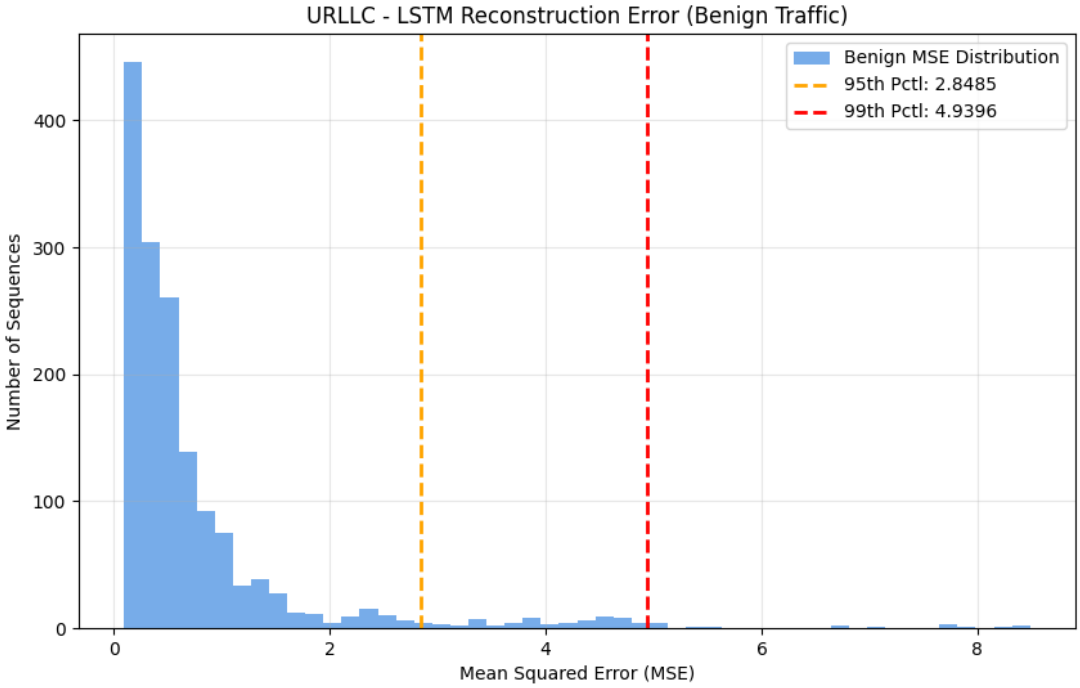
1. eMBB



2. mMTC



3. uRLLC


URLLC - LSTM Reconstruction Error (Benign Traffic)
Benign MSE Distribution
95th Pctl: 2.8485
99th Pctl: 4.9396
Number of Sequences
Mean Squared Error (MSE)
0
100
200
300
400
0
2
4
6
8

PERFORMANCE METRICS

1. eMBB

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  EVALUATING MODELS FOR: eMBB
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Loading mixed testing dataset...
Loading scaler...
Successfully aligned and scaled 37 features.

--- Evaluating One-Class SVM ---
/home/yashwanth-r/Capstone/NetworkSliceSecurityFramework/ML_Model/env/lib/python3.12/site-packages/sklearn/utils/validation.py:2691: Use
rWarning: X does not have valid feature names, but OneClassSVM was fitted with feature names
  warnings.warn(

SVM Confusion Matrix:
[[2967   56]
 [1832  953]]

SVM Classification Report:
              precision    recall  f1-score   support

      Benign       0.62      0.98      0.76      3023
   Malicious       0.94      0.34      0.50      2785

    accuracy                           0.67      5808
   macro avg       0.78      0.66      0.63      5808
weighted avg       0.77      0.67      0.64      5808


--- Evaluating LSTM Autoencoder ---
E0000 00:00:1776246786.893297   35214 cuda_platform.cc:52] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN E
RROR (303)

LSTM Confusion Matrix:
[[2863  151]
 [2784    0]]

LSTM Classification Report:
```

```
LSTM Classification Report:
              precision    recall  f1-score   support

      Benign       0.51      0.95      0.66      3014
   Malicious       0.00      0.00      0.00      2784

    accuracy                           0.49      5798
   macro avg       0.25      0.47      0.33      5798
weighted avg       0.26      0.49      0.34      5798
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2. mMTC

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  EVALUATING MODELS FOR: mMTC
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Loading mixed testing dataset...
Loading scaler...
Successfully aligned and scaled 34 features.

--- Evaluating One-Class SVM ---
/home/yashwanth-r/Capstone/NetworkSliceSecurityFramework/ML_Model/env/lib/python3.12/site-packages/sklearn/utils/validation.py:2691: Use
rWarning: X does not have valid feature names, but OneClassSVM was fitted with feature names
  warnings.warn(

SVM Confusion Matrix:
[[2404   58]
 [1635  518]]

SVM Classification Report:
              precision    recall  f1-score   support

      Benign       0.60      0.98      0.74      2462
   Malicious       0.90      0.24      0.38      2153

    accuracy                           0.63      4615
   macro avg       0.75      0.61      0.56      4615
weighted avg       0.74      0.63      0.57      4615


--- Evaluating LSTM Autoencoder ---

LSTM Confusion Matrix:
[[2239  214]
 [1163  989]]
```

```
LSTM Classification Report:
              precision    recall  f1-score   support

      Benign       0.66      0.91      0.76      2453
   Malicious       0.82      0.46      0.59      2152

    accuracy                           0.70      4605
   macro avg       0.74      0.69      0.68      4605
weighted avg       0.73      0.70      0.68      4605
```

3. uRLLC

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  EVALUATING MODELS FOR: URLLC
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Loading mixed testing dataset...
Loading scaler...
Successfully aligned and scaled 26 features.

--- Evaluating One-Class SVM ---
/home/yashwanth-r/Capstone/NetworkSliceSecurityFramework/ML_Model/env/lib/python3.12/site-packages/sklearn/utils/validation.py:2691: Use
rWarning: X does not have valid feature names, but OneClassSVM was fitted with feature names
  warnings.warn(

SVM Confusion Matrix:
[[1526   46]
 [1442 1019]]

SVM Classification Report:
              precision    recall  f1-score   support

      Benign       0.51      0.97      0.67      1572
   Malicious       0.96      0.41      0.58      2461

    accuracy                           0.63      4033
   macro avg       0.74      0.69      0.63      4033
weighted avg       0.78      0.63      0.61      4033


--- Evaluating LSTM Autoencoder ---

LSTM Confusion Matrix:
[[1484   79]
 [1848  612]]
```

```
LSTM Classification Report:
              precision    recall  f1-score   support

      Benign       0.45      0.95      0.61      1563
   Malicious       0.89      0.25      0.39      2460

    accuracy                           0.52      4023
   macro avg       0.67      0.60      0.50      4023
weighted avg       0.71      0.52      0.47      4023
```